

LEONARDO CHIMAL

309-278-3618 | leochi2565@gmail.com | East Moline, Illinois

<https://leonardochimal-portfolio-production.up.railway.app> | [linkedin.com/in/leonardo-chimal-a442b6267](https://www.linkedin.com/in/leonardo-chimal-a442b6267)

PROFILE

Engineering student with hands-on experience in CAD, robotics, mechanical design, embedded electronics, and software development. Experienced in taking systems from CAD and prototyping through hardware/software integration. Seeking an engineering internship in product design, prototyping, or electromechanical systems.

EDUCATION

Black Hawk College | Moline, Illinois | May 2027

Associate in Science (A.S.) | GPA - 3.36

United Township High School | East Moline, Illinois | May 2025

High School Diploma | GPA - 4.1 (unweighted)

EXPERIENCE

John Deere | Moline, Illinois | June 2024 – Present

Apprentice IT

- Developed internal tools and automation workflows using AWS, Terraform, JavaScript, and API integrations to support scalable backend systems and reduce manual processes
- Designed and deployed a secure URL shortening service with metadata tracking, encoding/decoding logic, and optimized backend performance for internal production use
- Built and integrated cloud-based services (AWS Lambda, SNS, SQS) using infrastructure-as-code to support event-driven system architecture and workflows
- Collaborated in Agile development environments using GitHub for version control, code reviews, and debugging, improving system reliability and deployment efficiency
- Mentored incoming IT apprentices by sharing onboarding knowledge, technical guidance, and best practices for internal systems and workflows.

First Tech Challenge | Moline, Illinois | May 2022 – 2025

Robotics Design Lead

- Led CAD design and mechanical integration for competitive FTC robotics systems using Onshape, collaborating with programming and manufacturing teams to optimize robot performance, maintainability, and weight distribution through iterative prototyping and engineering reviews.
- Designed drivetrain, intake, launcher, and structural robot systems while working alongside engineers and mentors from John Deere, Lockheed Martin, Meta, Tesla, Amazon, Black Hawk College, and other STEM organizations.
- Helped implement Agile/Scrum-style workflows using Trello, Discord task tracking, and sprint reviews while contributing to STEM outreach, mentoring, engineering presentations, and robotics demonstrations throughout the Quad Cities region.
- Qualified for the FTC World Championship in Houston, Texas (2024), FTC State Championship competitions (2025), and the FTC World Championship in Indianapolis, Indiana (2026).

PROJECTS

John Deere Enterprise URL Shortening Platform

URL: <https://leonardochimal-portfolio-production.up.railway.app/projects/url-shortener>

- Designed and implemented a production-grade internal enterprise URL shortening platform for John Deere, used daily by several hundred employees across the organization. Built with Next.js, TypeScript, MySQL, and a Rust-based redirect service, the system delivers device-aware routing, regex-driven URL handling, secure Okta SSO authentication, and a fully scalable AWS-powered analytics pipeline.

Languages: TypeScript, Rust, SQL

Development: Next.js, React, APIs, tRPC, Node.js

Databases/Tools: MySQL, Drizzle ORM, AWS (kinesis, S3, Athena), Docker, Terraform

Auth System: NextAuth, Okta SSO, Distributed Systems, Async Processing

Autonomous AI Drone Platform

URL: <https://leonardochimal-portfolio-production.up.railway.app/projects/drone>

- Designed and prototyped a modular 5-inch autonomous drone platform in Onshape, progressing from CAD to a powered bench-tested build. Integrated embedded electronics, high-current power distribution, vibration-aware mounting, and ELRS/Betaflight configuration while developing practical skills in ESC soldering, signal routing, and safe spin-up validation. Focused on iterative mechanical design, component integration, weight optimization, and preparing the platform for future autonomous expansion.

Languages: JavaScript, Python, SQL

CAD/Engineering: Onshape, CAD, 3D Printing, Drone Systems, Embedded

Development: React, Next.js, APIs, AWS